How do these definitions come from?

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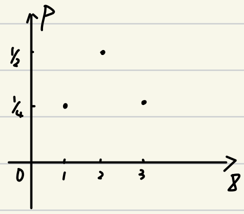
- Below, I shall give the intuition def of I_{series} $I_{discrete}$ and $I_{continuous}$

- Intuition Definition 1
  - suppose you want to store the info of a series in your computer disk, like the series of ↑↑↓↑↑↓↓··· of throwing a coin.
  - A good definition is $I \overset{?}{=} \log_2 \Omega$, where $\Omega = 2^N$
  - $I = N$, that is, you needs N bits to store N-throwing-coin

- Intuition Definition 2
  - For a discrete distribution $f(x)$, like



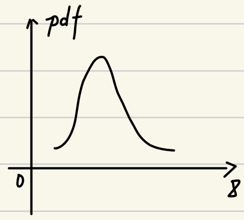
  - suppose you sample N times from it, $N \to +\infty$, then $\frac{I}{N}$ is a good measure of the info that a single sample contains, on average, which can be seen as the info of the distribution.

  - Suppose x has M values

$$I_{discrete} := \frac{I}{N} = \frac{1}{N}\log_2\Omega = \frac{1}{N}\frac{1}{\ln 2}\ln\Omega = \frac{1}{N}\frac{1}{\ln 2}\frac{N!}{N_1!\cdots N_M!} \overset{\text{Stirling}}{=} -\frac{1}{\ln 2}\sum_{i=1}^{M} P_i \ln P_i$$

We can also throw $\frac{1}{\ln 2}$, the $I_{discrete} = -\sum_i P_i \ln P_i$

- Intuition Definition 3
  - For a continuous distribution $f(x)$, like



  - $I_{discrete} = -\sum_i p(x_i)\Delta x \ln(p(x_i)\Delta x)$

$$= -\sum_i p(x_i)\Delta x\left(\ln p(x_i) + \ln\Delta x\right)$$

$$= -\sum_i p(x_i)\ln p(x_i)\Delta x + \ln\Delta x$$

$$\overset{\Delta x\to 0}{=} -\int p(x)\ln p(x)\,dx + \ln(\Delta x)$$

  - We define $I_{continuous} := -\int p(x)\ln p(x)\,dx$